

Interpaws Master Plan v3.0: The Agentic Shift

Status: Active | Phase: 3 (Agentic Intelligence)

Core Model: Qwen 3 8B (Instruct) | Architecture: ReAct (Reason + Act)

1. The Core Pivot

The previous "v2" plan focused on reactive features (answering questions based on text). **v3** pivots to **Agency**: the ability of the AI to perform multi-step reasoning and execute database queries autonomously.

Feature	v2 (Legacy)	v3 (Agentic)
Model	Llama 2 / 3 (Chat)	Qwen 3 8B
Logic	Retrieval Augmented Generation (RAG)	ReAct Loop (Thought → Tool → Observation)
Inventory	AI cannot see stock	AI queries <code>check_inventory</code> tool
Schedule	AI suggests "mornings"	AI checks real-time availability
Output	Text Paragraphs	JSON Actions + Natural Language

2. Technical Architecture

The "Brain": Agent Core (`backend/app/agent/core.py`)

Instead of a single prompt-response, the backend now implements a `while` loop that allows **Qwen 3 8B** to "think" and "act" up to 3 times per request.

- **System Prompt:** Strictly enforces **JSON Mode** for tool calls. Qwen 3 is specifically chosen for its superior instruction-following capabilities at the 8B size.

- **Safety:** The AI is "railroaded" into using specific, safe Python functions (Tools) to access data, preventing raw SQL injection.

The "Hands": Agent Tools ([backend/app/agent/tools.py](#))

The AI is given specific interfaces to interact with the PostgreSQL database:

1. **find_staff(query):** Uses vector search to find staff IDs based on complaint descriptions (e.g., "Limping dog" -> Orthopedic Surgeon).
2. **check_schedule(staff_id, date, time):** Wraps the existing [check_availability](#) logic to return [True/False](#) for specific slots.
3. **check_inventory(item_name):** Queries the [Medication](#) table to return stock levels (e.g., "Amoxicillin: 4 units").

3. Implementation Roadmap (Immediate)

Step 1: Infrastructure Upgrade

- **Task:** Switch Docker services to pull the **Qwen 3 8B** model.
- **Command:** [docker-compose exec ollama ollama pull qwen3:8b](#) (or [qwen2.5:7b](#) if [qwen3](#) tag is unavailable in specific registry).
- **Reason:** This model vastly outperforms previous open-source 8B models at following complex JSON instructions for tool usage.

Step 2: Backend Refactor

- **New Module:** [backend/app/agent/](#) containing [core.py](#) and [tools.py](#).
- **Endpoint:** [POST /agent/chat](#) replaces or augments the standard chat endpoint.
- **Logic:** The endpoint instantiates [InterpawsAgent](#) (powered by Qwen 3), runs the ReAct loop, and returns the final answer.

Step 3: Frontend Connection

- **Component:** [AIChat.js](#) updated to hit the new agent endpoint.
- **UX:** The UI remains the same, but the answers become factually grounded in the live database (e.g., "Yes, we have 3 slots open tomorrow" instead of "Please check the calendar").

4. Success Criteria

The system is considered "v3 Complete" when the following conversation is possible:

1. **User:** "My dog has a broken leg. Who can help?"
2. **Agent (Internal):** Calls [find_staff\("broken leg"\)](#) → Finds Dr. Smith (Orthopedic).
3. **Agent (Internal):** Calls [check_schedule\(Dr. Smith, Tomorrow\)](#) → Finds 2 PM slot.
4. **Agent (Reply):** "Dr. Smith specializes in orthopedics. She has an opening tomorrow at 2 PM. Shall I book it?"

